
EEG BIOMARKERS FOR ALZHEIMER'S

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PROBLEM DEFINITION

- **Alzheimer's disease (AD):** a progressive neurological disorder that affects memory and cognitive function.
- Traditional diagnostic methods are **costly, invasive, and time-consuming**. However, **EEG** provides a non-invasive and affordable way to monitor brain activity.
- Studies show AD patients experience:
 1. Increased delta and theta activity
 2. Decreased alpha and beta activity
- The project addresses **whether EEG brainwave patterns can effectively distinguish Alzheimer's patients from healthy individuals**.
- Importance:
 - Early detection may improve treatment planning
 - EEG-based analysis could support faster and more accessible diagnosis methods.
 - Machine learning can help automate and improve classification accuracy.

PROJECT OBJECTIVE

- Examine whether EEG-based features can serve as **reliable** indicators for identifying Alzheimer's disease.
- Compute relative power values for:
 - Delta
 - Theta
 - Alpha
 - Beta frequency bands
- Perform statistical analysis to identify differences between AD and HC groups.
- Apply machine learning models including: KNN, Logistic Regression, SVM, etc.

DATA PREPARATION

- Dataset consisted of EEG recordings from **9** Alzheimer's Disease patients and **9** Healthy Control subjects
- EEG signals were recorded using **16** EEG channels
- Sampling frequency: **256** Hz
- Recording durations ranged from: ~1.5 to 9 minutes per subject

Preprocessing Steps

- Raw EEG signals are high-dimensional and noisy.
- EEG signals were divided into standard frequency bands: Delta, Theta, Alpha, Beta
- Relative power for each band was calculated.

Final Prepared Dataset

- Each subject represented by a compact EEG feature vector.
- Labels assigned: AD (Alzheimer's Disease) and HC (Healthy Control)
- Prepared dataset used for Statistical testing and Machine learning classification models

MACHINE LEARNING MODELS

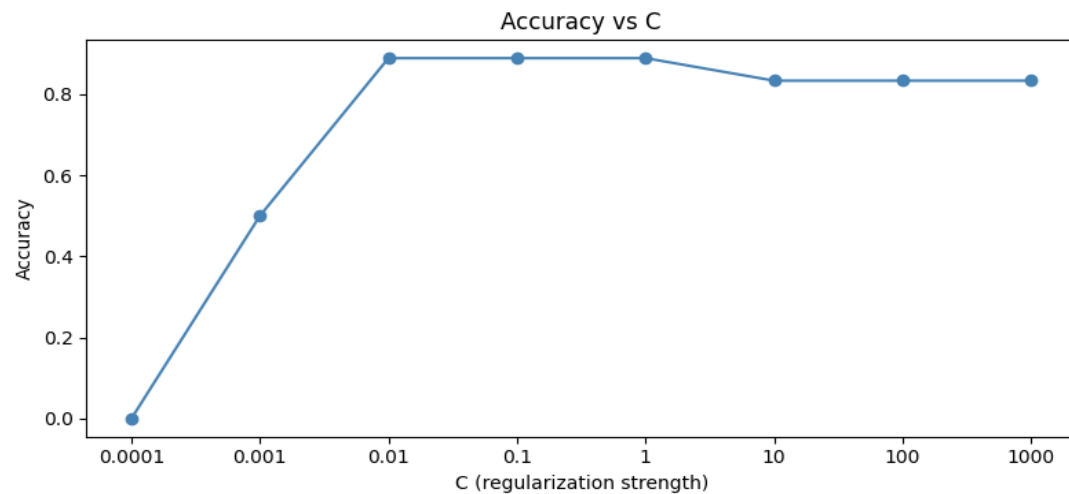
We trained 5 machine learning models:

- Logistic Regression: We chose this model because it works well with small datasets and is a simpler model.
- SVM: We chose this model because it is suited for high dimensional data and is suited for smaller datasets
- Random Forest: We chose this model to test whether a non-linear model can perform better on our dataset.
- Gradient Boosting: We chose this model to test if our dataset will perform in an iterative, error correcting approach
- KNN: We chose this model because it is effective when the feature space is divided into clusters.

LOGISTIC REGRESSION

Accuracy: 88.8%

Hyperparameter Tuning: Regularization Strength (C)

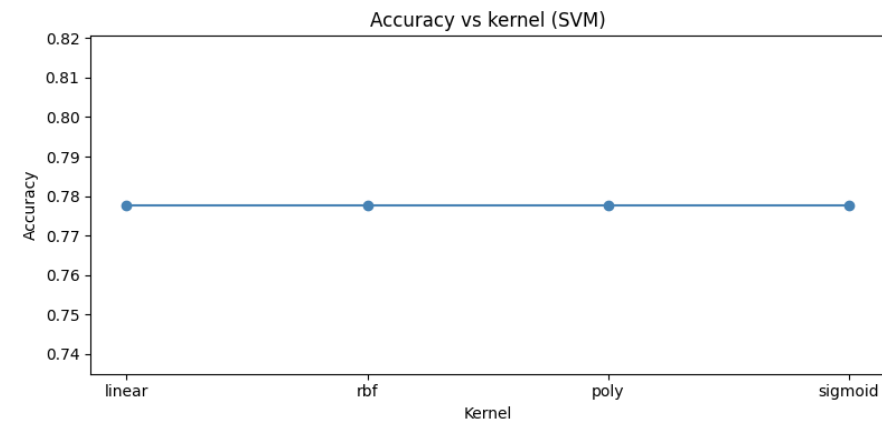
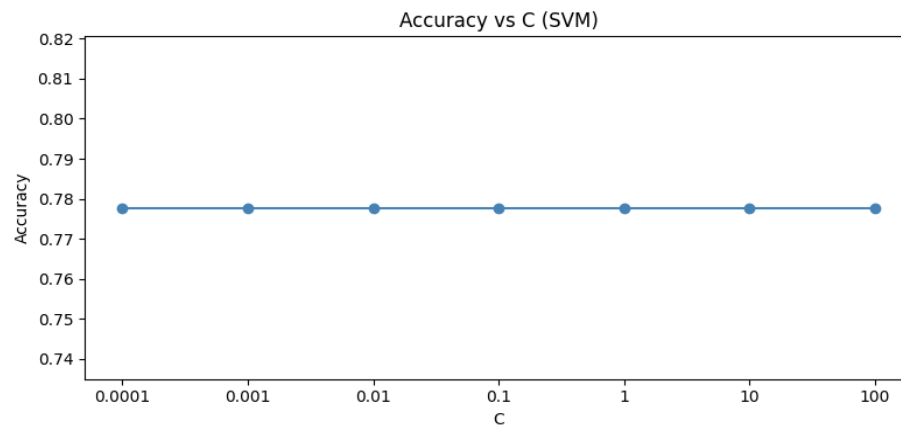


SVM

Accuracy: 77.7%

Hyperparameter Tuning:

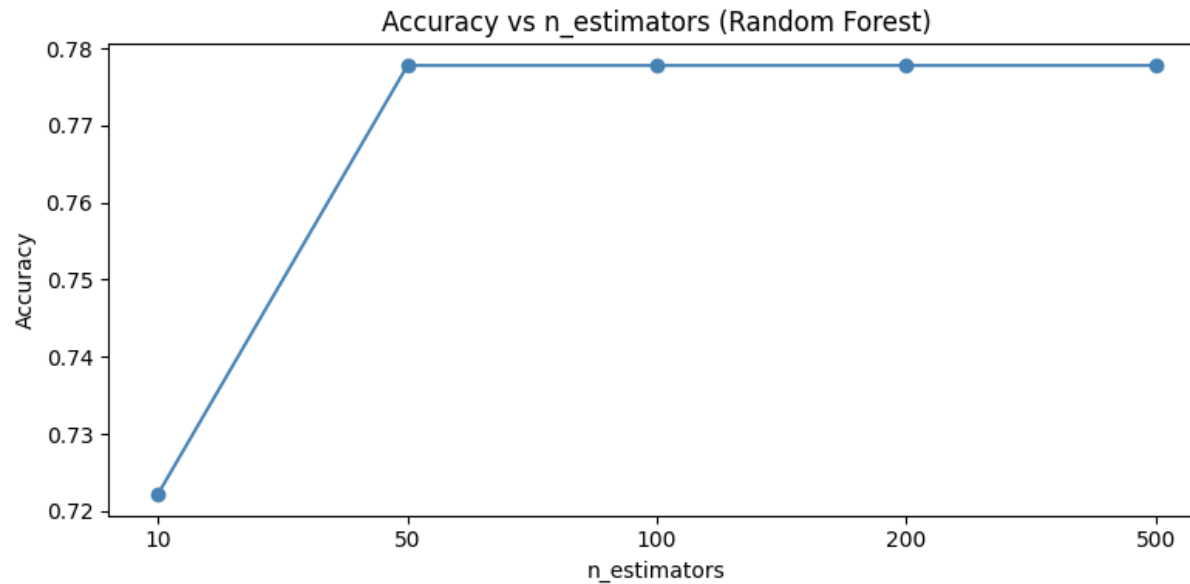
- Regularization Strength (C)
- Linear, RBF, polynomial, and sigmoid kernels



RANDOM FOREST

Accuracy: 77.7 %

Hyperparameter Tuning: N_estimators

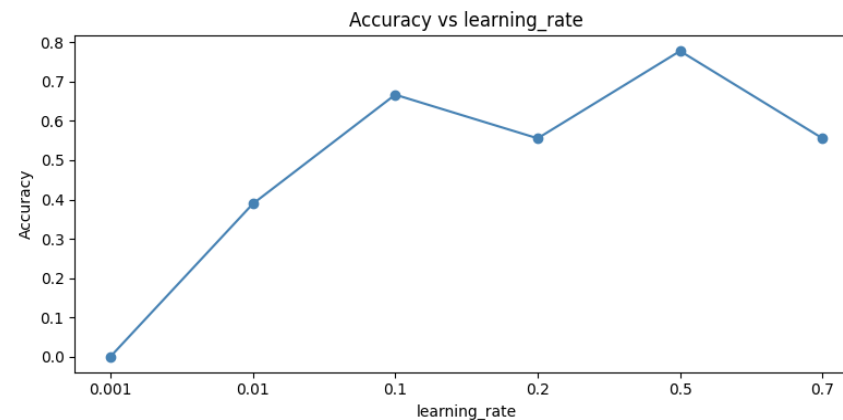
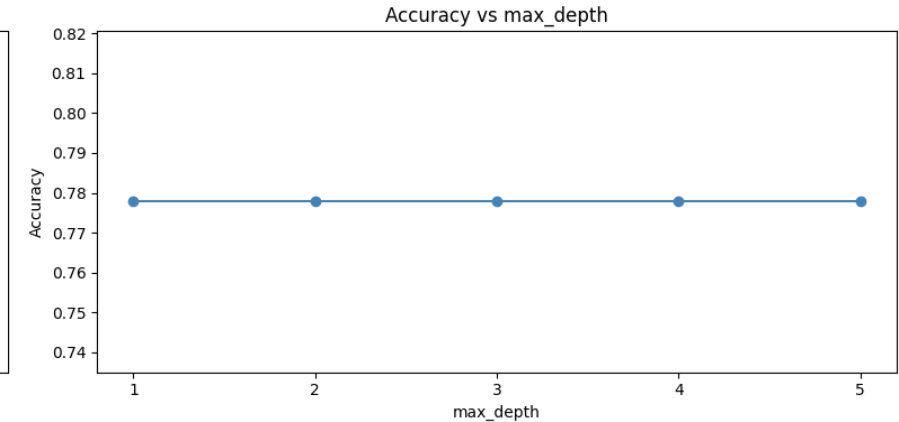
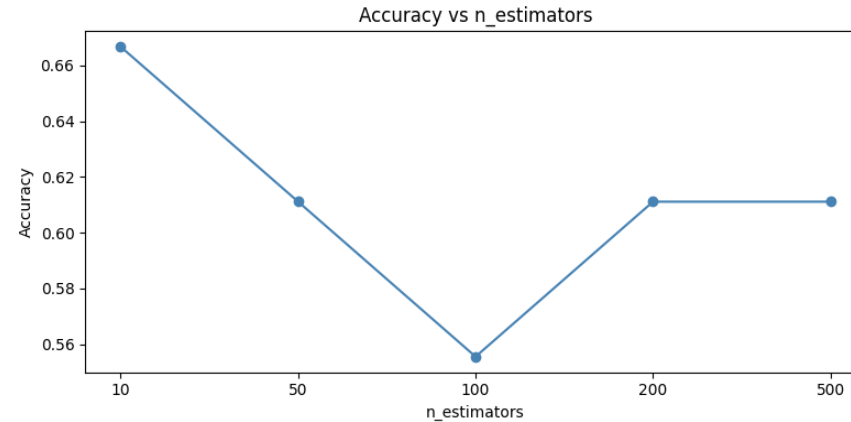


GRADIENT BOOSTING

Accuracy: 77.7%

Hyperparameter Tuning:

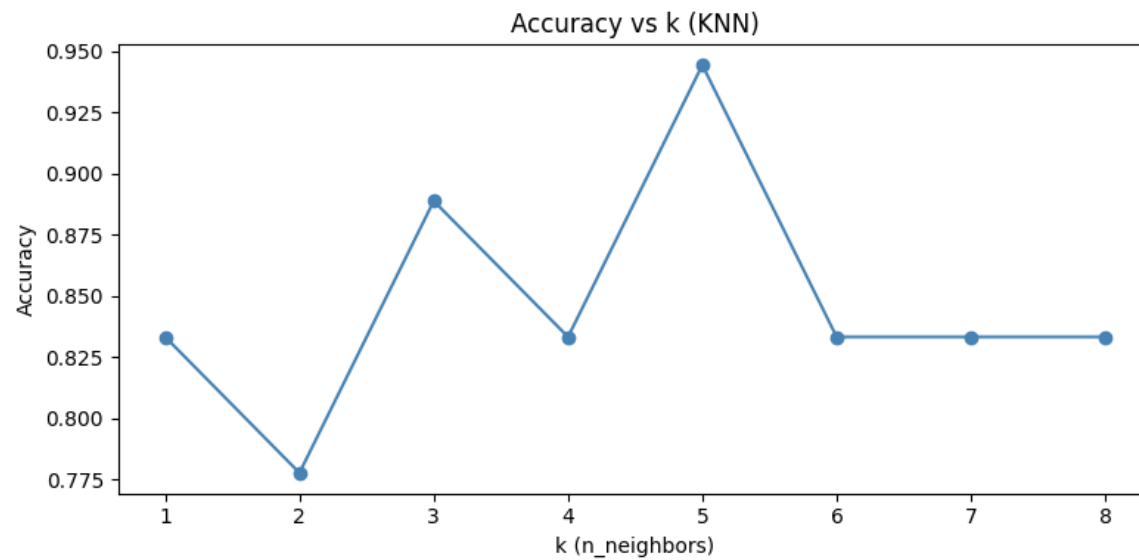
- N_estimators
- Learning Rate
- Max Depth



KNN

Accuracy: 94.4 %

Hyperparameter Tuning: N Neighbors



EXPERIMENTS + OBSERVATIONS

- Compared EEG band power features between AD and HC groups
- Theta power was significantly higher in AD subjects ($p < 0.01$)
- Alpha power was lower in AD subjects ($p \approx 0.053$)
- Results support the known EEG slowing effect in Alzheimer's disease
- AD subjects showed increased slow-wave activity (theta)
- HC subjects showed stronger alpha activity

EVALUATION METRICS

- Models evaluated using Leave-One-Out Cross Validation (LOO-CV)
- LOO-CV ensured every subject was tested exactly once
- Metrics used: Accuracy, Precision, Recall, F1 Score
- Confusion matrices used to analyze classification errors

KNN

- KNN achieved highest accuracy: 94.4%
- Correctly classified 17 out of 18 subjects
- Perfect recall for Alzheimer's cases
- Only one Healthy Control was misclassified
- Suggests EEG bandpower features form clear clusters in feature space

Classification Report:

	precision	recall	f1-score	support
HC	1.00	0.89	0.94	9
AD	0.90	1.00	0.95	9
accuracy			0.94	18
macro avg	0.95	0.94	0.94	18
weighted avg	0.95	0.94	0.94	18

Confusion Matrix:

```
[[8 1]  
[0 9]]
```

LOGISTIC REGRESSION

- Logistic Regression achieved a high classification accuracy of **88.9%**
- The model correctly classified **16 out of 18 subjects**.
- Performance was only slightly lower than KNN.
- The model demonstrated stable performance across both AD and HC groups

Classification Report:

	precision	recall	f1-score	support
HC	0.89	0.89	0.89	9
AD	0.89	0.89	0.89	9
accuracy			0.89	18
macro avg	0.89	0.89	0.89	18
weighted avg	0.89	0.89	0.89	18

Confusion Matrix:

```
[[8 1]
 [1 8]]
```

OTHER MODELS

- SVM and Random Forest both achieved an accuracy of 77.7%.
- Implementing complex kernels did not improve performance for the SVM model.
- Adding more trees did not show improvement

```
Classification Report:

              precision    recall  f1-score   support

      HC         0.78        0.78        0.78         9
      AD         0.78        0.78        0.78         9

 accuracy         0.78        0.78        0.78        18
 macro avg        0.78        0.78        0.78        18
weighted avg        0.78        0.78        0.78        18

Confusion Matrix:
[[7 2]
 [2 7]]
```

```
Classification Report:

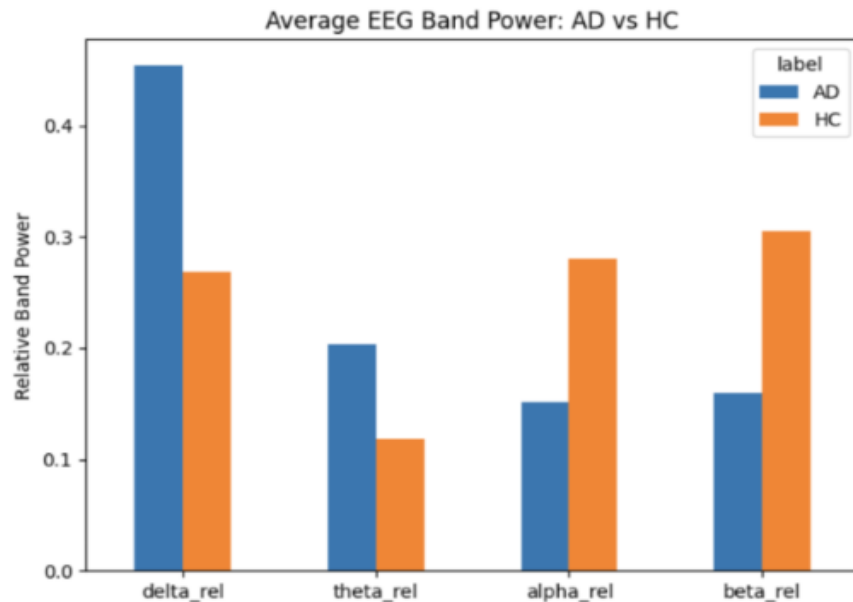
              precision    recall  f1-score   support

      HC         0.73        0.89        0.80         9
      AD         0.86        0.67        0.75         9

 accuracy         0.78        0.78        0.78        18
 macro avg        0.79        0.78        0.78        18
weighted avg        0.79        0.78        0.77        18

Confusion Matrix:
[[8 1]
 [3 6]]
```

RESULTS: FEATURE DIFFERENCES (AD VS. HC)



- AD subjects showed stronger slow-wave activity (delta and theta).
- HC subjects showed stronger fast-wave activity (alpha and beta).
- The observed frequency shift is consistent with EEG slowing in Alzheimer's disease.
- The separation between frequency patterns suggests EEG features contain distinguishable signals that can support machine learning classification.

Group	Delta Power	Theta Power	Alpha Power	Beta Power
AD	0.454	0.203	0.152	0.159
HC	0.269	0.119	0.280	0.305

RESULTS: THETA BAND ANALYSIS

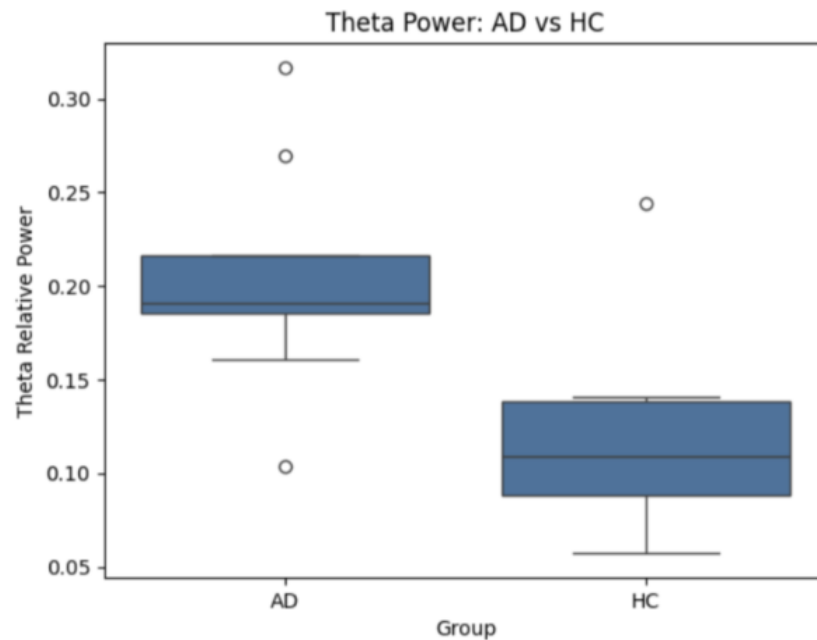
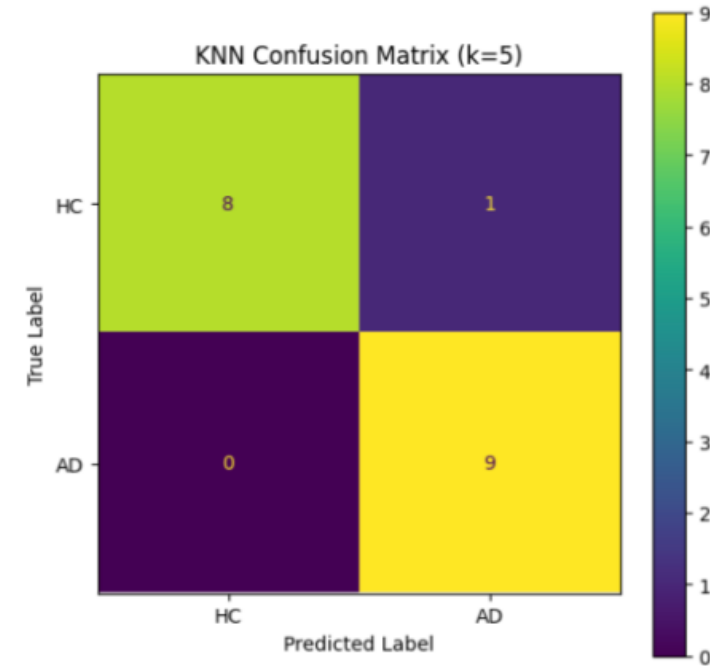
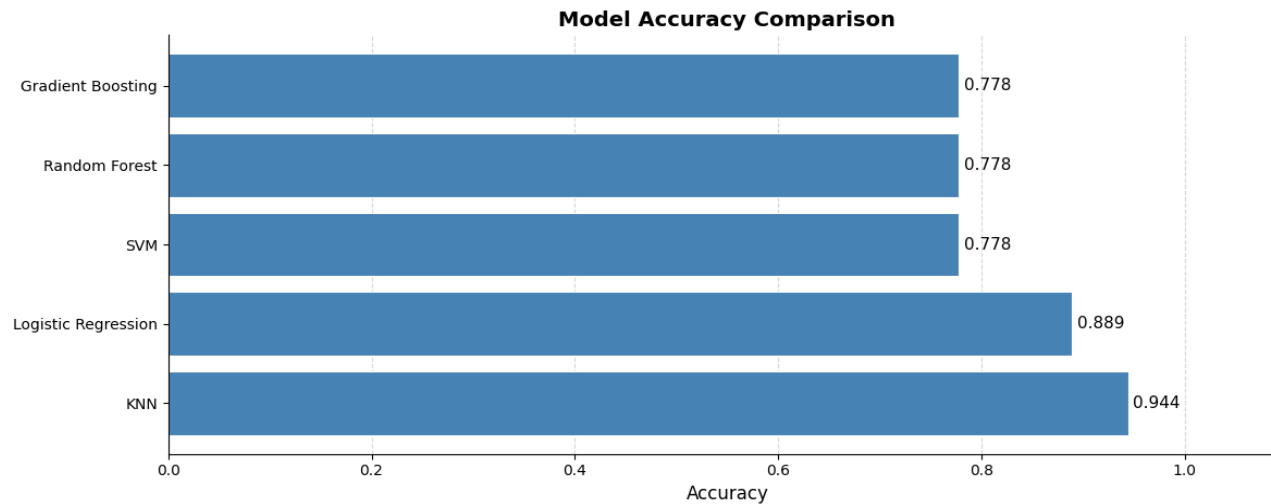


Figure 1. Boxplot of theta band power for Alzheimer's Disease (AD) and Healthy Control (HC) subjects.

- AD subjects consistently showed higher theta activity than HC subjects.
- Theta values exhibited relatively limited overlap between groups.
- Two-sample t-test showed statistical significance ($p = 0.0066$).
- Theta activity emerged as one of the strongest discriminative EEG features in the dataset.

RESULTS: MODEL INTERPRETATION



- KNN achieved the strongest overall classification performance.
- EEG feature patterns formed relatively distinguishable groupings between AD and HC subjects.
- Simpler models performed more effectively on the limited dataset.
- More complex models likely overfit due to the small sample size.

DISCUSSION

- EEG slowing may serve as a measurable biomarker for cognitive decline
- Strong performance from simpler models suggests the EEG features were relatively distinguishable and interpretable.
- Larger datasets are needed to evaluate whether these patterns generalize reliably.
- More complex models may require substantially more EEG data to learn stable patterns.
- EEG-based ML approaches could support earlier and more accessible Alzheimer's screening in the future.

FUTURE WORK -DATA

- Dataset Improvements
 - Increase dataset size beyond 18 subjects
 - Use more diverse EEG recordings
 - Improve generalization and reduce variability
- Feature Engineering
 - Move beyond relative EEG band power
 - Explore richer features:
 - Wavelet transforms (time–frequency info)
 - Spectral entropy (signal complexity)
 - Functional connectivity (brain network activity)
 - Include spatial relationships between EEG electrodes

FUTURE WORK - MODEL

- Model Optimization
 - Use systematic hyperparameter tuning (Grid / Random Search)
 - Improve performance of KNN (best model)
 - Ensure fair comparison across all models
- Advanced & Clinical Work
 - Explore deep learning models (CNNs, RNNs)
 - Extend to Mild Cognitive Impairment (MCI)
 - Predict disease progression for early diagnosis